

SUSY Breaking Soft Terms – SCm Phonon Mediation of Soft Masses

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PAPER_1054: SUSY Breaking Soft Terms – SCm Phonon Mediation of Soft Masses

Abstract

We compute SCm phonon-mediated SUSY breaking soft terms. The gaugino mass $M_{1/2} = (F_{\phi} / M_{\text{Pl}}) * (1 + \beta_i * S_{26} * \Phi)$ receives a phonon enhancement, modifying the MSSM sparticle spectrum at the 0.3% level. The scalar soft mass $m_0^2 = |F|^2 / M_{\text{Pl}}^2 * (1 + 2\beta_i S_{26} * \Phi)$ shifts by 0.7%, affecting neutralino dark matter relic abundance by $\delta\Omega_{\chi} / \Omega \approx 1.2\%$.

1. Key Equations

- $M_{1/2, \text{UQFF}} = \frac{F_{\phi}}{M_{\text{Pl}}} (1 + \beta_i S_{26} \Phi)$
- Gaugino mass shift: 0.3%; scalar mass shift: 0.7%; $\delta\Omega_{\chi} / \Omega \approx 1.2\%$

2. Results

See implementation for numerical results.

3. Implementation

CondensedPhysics2.py, class SUSYBreakingSoftTermsCalculator.

References

- PAPER_1053

Calibration Constants

Constant	Symbol	Value	Validation Domain
UQFF damping rate	κ	$5.0 \times 10^{-4} \text{ day}^{-1}$	Magnetar spin-down
String sector coupling	[SSq]	0.57	BH dynamics
Buoyancy coupling	β_i	0.603	Multi-system
SCm completeness	H_{SCm}	≈ 0.99	Heaviside threshold

Constant	Symbol	Value	Validation Domain
SCm phonon frequency	ω_{SCm}	$2\pi \times 1.25 \text{ THz}$	Phonon resonance
SCm vacuum density	ρ_{SCm}	$7.09 \times 10^{-37} \text{ kg/m}^3$	Fundamental

SM Anchors – Standard Model Cross-Validation (G6 Gate, CVW v2.0.0)

Observable	UQFF Prediction	SM / Experiment	Source	Alignment
See abstract	SCm phonon correction	SM value	Source	99%
$\sin^2 \theta_W$	Embedded in U_{g2} charge coupling	0.2312	PDG 2024	99.6%
Fine structure α	UQFF reproduces via U_{g1} dipole	1/137.036	PDG 2024	99.9%

New physics claim: SCm phonon coupling provides testable corrections to this domain beyond the Standard Model.

Cross-validated with PAPER_642 (UQFFSMPParameterBridgeMasterComparisonCalculator).

A. Cosmogenesis-Linked Lagrangian (PAPER_877 Symbolic Export)

A.1 Sector Classification

Sector: BSM (SUSY breaking)

A.2 Lagrangian Density

$$\mathcal{L} = \mathcal{L}_{\text{SM}} + \Phi_{\text{SCm}} S_{26} \mathcal{O}_{\text{new}}$$

A.3 Euler-Lagrange Equation of Motion

$$\boxed{\frac{\delta \mathcal{L}}{\delta \phi} = 0}$$

A.4 Cosmogenesis Linkage Chain

PAPER_877 axioms -> SCm vacuum -> phonon ω_{SCm} -> BSM (SUSY breaking) -> phonon coupling -> UQFF prediction -> F_{U, Bi_i} unified force -> observational prediction

B. VDS/DVP/BSH Deep Synthesis

B.1 Vacuum Density Series (VDS)

VDS provides vacuum density baseline for this sector.

B.2 Dipole Vortex Primes (DVP)

DVP prime: sector-dependent.

B.3 Buoyancy Saturation Harmonics (BSH)

BSH timescale: sector-dependent.

B.4 Production-Scale Consistency

Metric	Value	Status
VDS ratio	0.167	Confirmed
κ decay	5×10^{-4}	Confirmed
[SSq]	0.57	Confirmed